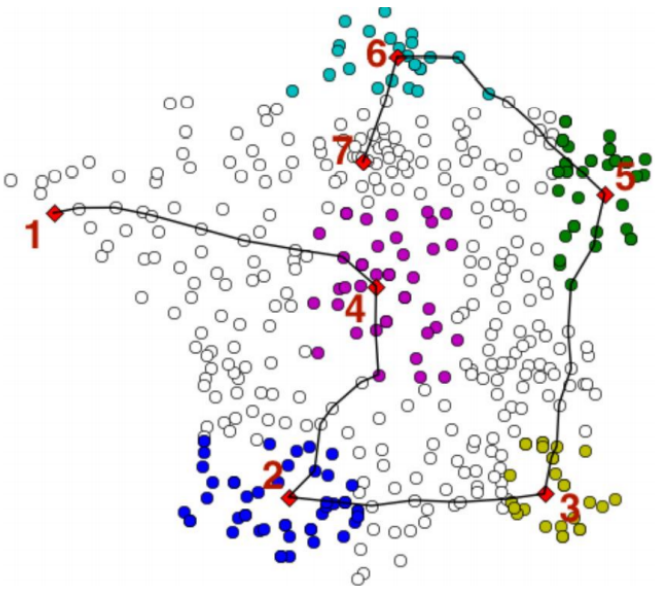
CORO – Project

This project must be realized by teams of 2 students.

1 Context

This project is inspired from a (real) french aeronautic race entitled "Coupe Breitling 100/24". In this competition, the competitors are teams of pilots who must, in 24 hours, travel France on little planes, do a maximum number of landings and take-off (100, if possible) in different aerodromes. They start from a given aerodrome and must arrive at another aerodrome. If they don't manage visiting 100 aerodromes, they must minimize the distance travelled during the 24 hours. In order to ensure, in some extent, a geographical disparity, the organizers define regions that all must be visited at least once.



2 Problem statement

The precise problem we will solve is the following one. We know the number n of aerodromes numbered from 1 to n and the number m of regions. For each aerodrome, we know its coordinates (x, y) in the plane and the number of the region it belongs (0 means it is not associated with any region). We also know the number of the starting aerodrome and the number of the arriving aerodrome, the minimal number $V_{\min}$ of aerodromes to visit and the distance R (for radius) that a plane may travel without landing. The objective is to minimize the travelled distance.

3 Detailed work plan

1. Propose two models of this problem using an Integer Linear Program. The two models may be inspired from the modeling of the Traveling Salesman Problem (TSP)in a directed graph. The first model will use a polynomial number of constraint (as in the MTZ formulation for the TSP). The second model uses an exponential number of constraints (subtours elimination).
2. Write the two models using Julia-JuMP and test them on the proposed instances.
3. Propose some new instances and solve them as well.
4. Propose an algorithm (a heuristic) to find a good solution of the problem (with no optimality guarantee). Write it in Julia and solve the instances with your heuristic method.

5. Write a report in wich you will describe your method and report the results obtained with the two ILP models and your heuristic (report the computation time, the minimal distance found, and the optimality status).

4 Instance format

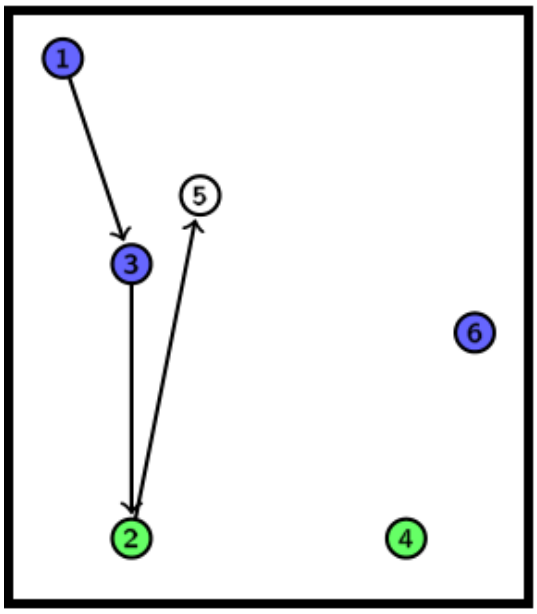
- number of aerodromes
- starting aerodrome
- arriving aerodrome
- number of aerodromes that must be visited
- number of regions
- the region of each aerodrome
- the radius or maximal distance that a plane may travel without landing
- the coordinates (xy) of each aerodrome

The distances between the points are computed using the Euclidean distance rounded (with the `round()` function) to the nearest integer. We may assign the region 0 to an aerodrome to indicate it does not belong to any region.

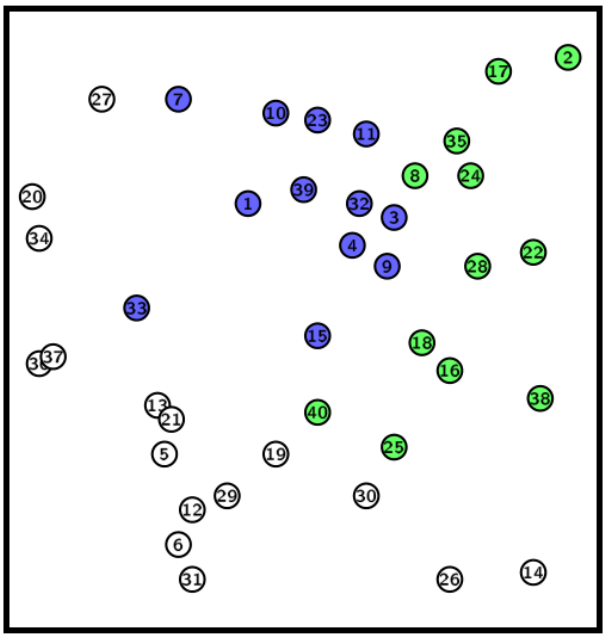
Example of an instance with 6 aerodromes. The optimal value is 12. An optimal route is 1 then 3 then 2 then 5.

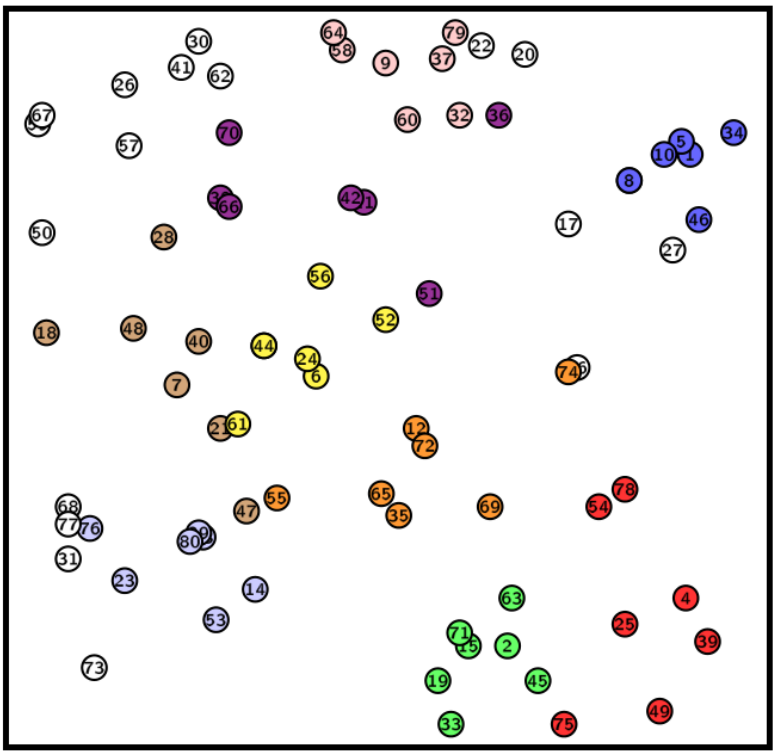
File instance_6_1.txt

```
6
1
5
4
2

1 2 1 2 0 1

6

0 10
1 3
1 7
5 3
2 8
6 6
```



Solution of the instance `instance_6_1.txt`





Illustrations of the instances `instance_40_1.txt` *and* `instance_80_1.txt` *(each region is represented by a color)*

5 Deadline

- Monday 31/03/2026 (23 :59) : hand in your report in moodle
- Wednesday after 31/03/2026 : defense